

Sean Luka Girgis

Python / Unix Shell Automation Developer | Data Feeds, Validation & Production Support

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Professional Summary

Senior technology professional with 20+ years of enterprise experience across Python, Unix/Linux, KornShell/ksh, Perl, SQL, data feed handling, automation, validation, logging, monitoring, performance troubleshooting, and production support. Strong background designing repeatable scripts and data workflows that extract, validate, transform, monitor, and report operational data across large-scale enterprise systems. Best aligned to Python and Unix shell scripting roles requiring task automation, data feed handling, business logic implementation, diagnostics, performance tuning, reliability, Git/version control, Docker familiarity, and cross-functional delivery.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas automation pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational feeds into validated reporting, analytics, and forecasting datasets.
- Developed Python scripts for data extraction, cleansing, validation, reconciliation, reporting, and repeatable operational processing across monitoring and capacity-management data sources.
- Implemented validation checks and diagnostic workflows to identify missing feeds, inconsistent metrics, schema drift symptoms, transformation issues, and downstream reporting discrepancies.
- Designed logging, monitoring, and troubleshooting practices around data workflows to support issue diagnostics, performance tracking, and production reliability.
- Optimized SQL and Python-driven data processing workflows for performance, scalability, maintainability, and reliable execution across large enterprise datasets.
- Worked with Linux/Unix-style automation patterns, Git, Docker/containerized Python jobs, operational runbooks, and cross-functional production support processes.
- Documented transformation rules, validation checkpoints, support procedures, metric definitions, and troubleshooting steps to make recurring operational work easier to maintain.
- Collaborated with infrastructure, application, analytics, and business teams to translate operational requirements into practical automation, reporting, and support solutions.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Created operational analysis workflows that correlated synthetic monitoring results, real-user performance signals, and application behavior to support performance improvements.
- Built dashboards and reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported production issue analysis by tracing monitoring signals, transaction behavior, and system performance data.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, dashboards, operational analytics, and production reliability.
- Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed reusable scripts, dashboards, alert logic, SLA thresholds, technical documentation, and operational standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze telemetry, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and data feed processes for telecom-scale transaction systems.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.
- Implemented business logic in billing and integration workflows using C/C++, Pro*C, PL/SQL, SQL scripts, XML, shell automation, and enterprise messaging interfaces.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, KornShell/ksh, Bash concepts, Perl, Awk, SQL, PL/SQL, XML, JSON, C++, C, Java

Flagship Projects

AI-Powered Job Search Pipeline

Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking.

Emphasized repeatable scripting, validation, structured outputs, and operational reliability.

Technologies: Python, JSON, FAISS, Sentence Transformers, Grok / xAI API, Quality Gates, Automation

HorizonScale — Capacity Forecasting Automation

Built Python/PySpark automation workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting and dashboard workflows.

Technologies: Python, PySpark, SQL, Pandas, Prophet, scikit-learn, Streamlit

Enterprise APM Reporting Automation

Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.

Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Automation, Monitoring